

Solutions: Calculus

ECON 441: Introduction to Mathematical Economics

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Exercise 6.2

2. $y = 5x^2 - 4x$

(a)

$$\begin{aligned}\frac{\Delta y}{\Delta x} &= \frac{f(x + \Delta x) - f(x)}{\Delta x} \\ &= \frac{5(x + \Delta x)^2 - 4(x + \Delta x) - (5x^2 - 4x)}{\Delta x} \\ &= \frac{5(x^2 + 2x\Delta x + (\Delta x)^2) - 4x - 4\Delta x - 5x^2 + 4x}{\Delta x} \\ &= \frac{10x\Delta x + 5(\Delta x)^2 - 4\Delta x}{\Delta x} = 10x - 4 + 5\Delta x\end{aligned}$$

(b)

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} (10x - 4 + 5\Delta x) = 10x - 4$$

(c) $f'(2) = 10 \times 2 - 4 = 16$ and $f'(3) = 10 \times 3 - 4 = 26$.

3. $y = 5x - 2$

(a)

$$\frac{\Delta y}{\Delta x} = \frac{5(x + \Delta x) - 2 - (5x - 2)}{\Delta x} = \frac{5\Delta x}{\Delta x} = 5$$

It is a constant function.

(b) No: the difference quotient equals 5 whatever the size of Δx , so its limit as $\Delta x \rightarrow 0$ is also 5, i.e. $dy/dx = 5$.